

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EE)

April - May 2013

EE310 Programmable Logic Controller and Industrial Automation

Date: 04.05.2013, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) State the difference between sequential/logical and analog/automatic control [07]

Q - 2 (a) State the advantages of PLC [04]

(b) Explain in detail butterfly valve [05]

(c) Define control valves and state the detailed classification of control valves [05]

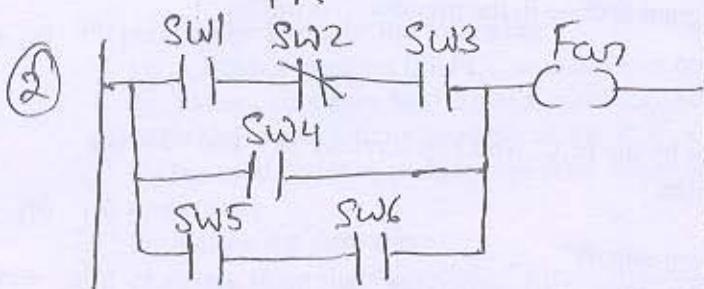
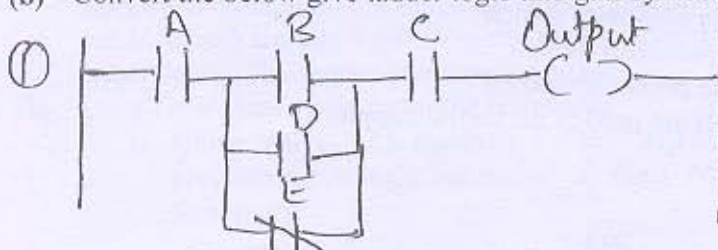
OR

Q - 2 (a) Prepare ladder logic for the following: [10]

- 1) There is an alarm system which has 4 hazard inputs which go on as some operational malfunction occurs. The system operates as follows: If one input is on, nothing happens. If any two inputs are on, a red pilot light goes on. If any 3 inputs are on, an alarm siren sounds. If all 4 inputs are on, the fire department is notified.

- 2) 4 motors M1, M2, M3 and M4 are required to ON in sequential way one after other after some time interval. In this process, the sequence should not stop at M4 but the motor M1 starts once again (after M4 off) and sequence run continuously.

(b) Convert the below give ladder logic into gate symbol and statement: [04]



Q - 3 (a) For the below given application which sensors will be used for measurement: [04]

- 1) There is a furnace whose temperature remains in the range of 500°C to 600°C.
- 2) In certain plant the water after being used in the process is being disposed off. We need to measure the flow with which the waste water is exiting through

open drain.

- 3) There is a pipe from which slurry flows. Slurry is conductive. We need to measure the flow of slurry through pipe.
- 4) There is a 1m high tank which has conductive fluid present in it. We need to find the liquid level in tank.
- (b) Explain in detail Modbus protocol [05]
- (c) Explain in detail Piezoelectric sensor. Also discuss which physical parameters can be measured using it giving few examples for each [05]

OR

Q - 3 (a) Multiple Choice questions: [10]

- 1) PLC is based in the fundamentals of
 - i. Analog Electronics
 - ii. Microelectronics
 - iii. Logic and Boolean Algebra
- 2) LOGIC means:
 - i. Decision making Element
 - ii. Related with 1 and 0
 - iii. Combinational Circuits
- 3) The opto-coupler used in PLC input unit is for
 - i. To isolate input component with CPU
 - ii. For switching function
 - iii. None of these
- 4) The IEC standard defines the number of programming language for PLC
 - i. 2
 - ii. 3
 - iii. 5
- 5) A thermistor is a
 - i. Positive temperature coefficient device
 - ii. Based on Seebeck effect
 - iii. Negative temperature coefficient device
- 6) Opaque surface can be sensed by
 - i. Inductive type proximity switch
 - ii. Optical type proximity switch
 - iii. Magnetic type proximity switch
- 7) Ni-Cr or Ni-Al metals are used in the thermocouple
 - i. J-type
 - ii. PT-type
 - iii. K-type
- 8) Simulation of the program is done in the process
 - i. Specification
 - ii. Realization
 - iii. Design
- 9) The power supply unit by the PLC, which generates +24V and +5V is a
 - i. Ordinary rectifier
 - ii. Battery
 - iii. Regulated power supply
- 10) A counter in a ladder rung after satisfying the input contacts true, will energized its contact after
 - i. Any few seconds
 - ii. Generating number of pulses
 - iii. Counting number of pulses

- (b) Explain in detail output module present in PLC's block diagram

[04]

SECTION - II

- Q - 4 (a) Prepare ladder logic for the following;

[07]

- 1) There are 4 car spaces in a parking. Car is to be detected and allowed to enter into the parking place if space is available. If no space, full indication lamp should be light, otherwise indication lamp should light to indicate the proper space. (Do not place master start/stop button)
- 2) For 30 seconds count the pulses from the sensor and then operate an output O1.

- Q - 5 (a) A process has 4 inputs which are connected to 4 PLC counter:

[07]

A	Up Counter
B	Down Counter (preset at 20)
C	Up Counter
D	Down Counter (Preset at 25)

When A reaches 7 or B reaches 0, output L is to go on.

When C reaches 6 and D reaches 0, output M is to go on.

When both L and M are on, Output N is to go on

Process master ON and RESET to be used.

- (b) Design the ladder diagram for controlling the traffic light (make necessary assumption for timing lights and prepare ladder only for one set of Red, green and yellow lights)

[07]

OR

- Q - 5 (a) Write a note on factors considered while selecting a PLC

[04]

- (b) Explain in detail factors to be considered to ensure continuous, reliable operation of the PLC system after installation.

[05]

- (c) List out the things done or followed for maintenance of a PLC system

[05]

- Q - 6 (a) Write a ladder diagram program for a system of 3 conveyors. Items placed on 1st and 2nd conveyors are detected by the photocell and counted. When the 6 items are counted on the 1st conveyor and 8 items are counted on the 2nd conveyor then both are stopped and the main conveyor is started. After some time interval when whole of the batch (items) are taken out from the main conveyor, it also stops and indicated by an output lamp. The sequence is running again.

[07]

- (b) Draw flow sheet symbols for the following:

[07]

- 1) Globe Valve
- 2) Butterfly Valve
- 3) Four Way Valve
- 4) Field mounted programmable logic controller
- 5) Horn
- 6) Flow nozzle
- 7) Magnetic type flow meter

OR

- Q - 6 (a) Prepare ladder logic for the following:

[05]

- 1) A light is to go on if a PLC counter does not have values of either 23 or 31.
- 2) Three conveyors feed a main conveyor. The count from each feeder conveyor is fed into an input variable in the PLC. Construct a PLC program to obtain the total count of parts on the main conveyor.

- (b) Do as directed:

[03]

- 1) Define the term relay
- 2) Draw flow sheet symbol of Electromagnetic or sonic signal (both guided and non guided)
- 3) Define the term shared controller.

- (c) Explain in detail field bus and control net protocols

[06]
